

Sizhuo Ma

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Personal website • Google Scholar • LinkedIn

Summary

- Research interests: image/video restoration and enhancement, generative AI, computational imaging, single-photon imaging, computer vision, machine learning.

Education

University of Wisconsin–Madison

Dec. 2016 – Jan. 2022

Ph.D. in Computer Sciences

Thesis: Resolving Motion with Single-Photon Cameras

University of Wisconsin–Madison

Aug. 2014 – Dec. 2016

M.S. in Computer Sciences

GPA: 3.92/4.00

Shanghai Jiao Tong University, China

Sep. 2010 – Jul. 2014

B.S. in Computer Science and Engineering

GPA: 90.3/100

Work Experience

Snap Inc.

Feb. 2022 – Present

Senior Research Scientist

- Develop and deploy spatial/temporal frame upsamplers for **efficient video generation**.
- Design visual token pruning methods that reduce the inference cost of **vision-language models**.
- Build server-side and on-device **video super-resolution** models that improve video quality and user engagement.
- Train video **engagement prediction** models for large-scale content understanding and recommendation.
- Develop on-device **face image quality assessment** models for real-time visual quality evaluation.
- Create **computational imaging and display** technologies that improve user experience.
- **Mentor** research interns and publish research in top-tier computer vision and graphics venues.

WISION Lab, University of Wisconsin–Madison

May. 2016 – Jan. 2022

Graduate Research Assistant (Advisor: Prof. Mohit Gupta)

- Developed novel solutions to motion-related computer vision problems (*e.g.*, scene flow, burst photography) with computational camera designs (*e.g.*, light field, structured light, single-photon cameras).

Snap Inc.

May. 2020 – Aug. 2020

Research Intern, Computational Imaging Team (Supervisor: Shree Nayar)

- Developed computational methods that significantly improve the decoding distance of QR codes (4×).

Living Environments Lab, University of Wisconsin–Madison

Jan. 2016 – May. 2016

Graduate Research Assistant (Advisor: Prof. Kevin Ponto)

- Built mobile AR prototypes with Unity, OpenCV and Caffe.

Visual Media and Data Management Lab, Shanghai Jiao Tong University

Sep. 2012 – Jun. 2014

Undergraduate Research Assistant (Advisor: Prof. Bin Sheng)

- Implemented a real-time, monocular, dense SLAM system in C++ as a platform for AR applications.

Teaching Experience

Teaching Assistant

Sep. 2015 – Jan. 2016

CS301: Introduction to Data Programming (Python), University of Wisconsin–Madison

Teaching Assistant

Sep. 2014 – May. 2015

CS302: Introduction to Programming (Java), University of Wisconsin–Madison

Publications

- [1] Guohao Sun, Yufei Wang, **Sizhuo Ma**, Yuege Xie, Yuting Cheng, Zhiqiang Tao, Jian Wang. IF-Prune: Information-Flow Guided Token Pruning for Efficient Vision-Language Models. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026)*
- [2] Zhihang Zhong, Gurunandan Krishnan, Wei Wang, Xiao Sun, Yu Qiao, **Sizhuo Ma***, Jian Wang*. Velocity Disambiguation for Video Frame Interpolation. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI 2026)*
- [3] Ayush Garg, **Sizhuo Ma**, Mohit Gupta. gQIR: Generative Quanta Image Reconstruction. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026)*
- [4] Eric Ming Chen, Di Liu, **Sizhuo Ma**, Michael Vasilkovsky, Bing Zhou, Qiang Gao, Wenzhou Wang, Jiahao Luo, Dimitris N. Metaxas, Vincent Sitzmann, Jian Wang. Snapmoji: Instant Generation of Animatable Dual-Stylized Avatars. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2026)*
- [5] Jiahao Guo, Yifan Ji, Zhenzhong Chen, Yufei Wang, **Sizhuo Ma**, Yuwei Guo, Yulun Zhang, Jian Wang. Towards Redundancy Reduction in Diffusion Models for Efficient Video Super-Resolution. *arXiv, 2025*
- [6] **Sizhuo Ma**, Wei-Ting Chen, Qiang Gao, Jian Wang, Chris Wei Zhou, Wei Sun, Weixia Zhang, Linhan Cao, *et al.* VQualA 2025 Challenge on Face Image Quality Assessment: Methods and Results. *ICCV Workshop 2025*
- [7] Dasong Li*, **Sizhuo Ma***, Hang Hua*, Wenjie Li*, Jian Wang*, Chris Wei Zhou*, Fengbin Guan, Xin Li, *et al.* VQualA 2025 Challenge on Engagement Prediction for Short Videos: Methods and Results. *ICCV Workshop 2025*
- [8] **Sizhuo Ma**, Karl Bayer, Gurunandan Krishnan, Mohit Gupta, Shree Nayar. Privacy-Enabled Parallax Display. *IEEE Conference Virtual Reality and 3D User Interfaces (VR 2025)*
- [9] Howard Zhang+, Yuval Alaluf+, **Sizhuo Ma**, Achuta Kadambi, Jian Wang*, Kfir Aberman*. InstantRestore: Single-Step Personalized Face Restoration with Shared-Image Attention. *SIGGRAPH 2025*
- [10] Pradyumna Chari, **Sizhuo Ma**, Daniil Ostashev, Achuta Kadambi, Gurunandan Krishnan, Jian Wang*, Kfir Aberman. Personalized Restoration via Dual-Pivot Tuning. *IEEE Transactions on Image Processing (TIP 2025)*
- [11] Junho Kim, Young Min Kim, Ramzi Zahreddine, Weston A. Welge, Gurunandan Krishnan, **Sizhuo Ma***, Jian Wang*. Privacy-Preserving Visual Localization with Event Cameras. *IEEE Transactions on Image Processing (TIP 2025)*
- [12] Baiang Li, **Sizhuo Ma**, Yanhong Zeng, Xiaogang Xu, Youqing Fang, Zhao Zhang, Jian Wang*+, Kai Chen*. Exposure-Limited Image Enhancement with Generative Diffusion Prior. *International Conference on Computational Photography (ICCP 2025)*
- [13] Yiming Zhang, Lionel Zhe Wang+, **Sizhuo Ma+**, Xinjie Li, Jian Ren, Zhihang Zhong*, Jian Wang*. DiffBody: Human Body Image Restoration with Generative Diffusion Prior. *International Conference on Computational Photography (ICCP 2025)*
- [14] Aditya Arora, Zhengzhong Tu+, Yufei Wang+, Ruizheng Bai, Jian Wang*, **Sizhuo Ma***. GuideSR: Rethinking Guidance for One-Step High-Fidelity Diffusion-Based Super-Resolution. *arXiv, 2025*
- [15] Wei-Ting Chen+, Vong Yu Jiet+, Yi-Tsung Lee, Qiang Gao, Sy-Yen Kuo, **Sizhuo Ma***, Jian Wang*. DiffVQA: Video Quality Assessment Using Diffusion Feature Extractor. *arXiv, 2025*
- [16] Jian Wang, **Sizhuo Ma**, Karl Bayer, Yi Zhang, Peihao Wang, Bing Zhou, Shree Nayar, Gurunandan Krishnan. Perspective-Aligned AR Mirror with Under-Display Camera. *SIGGRAPH Asia 2024 (Journal, Best Paper Award)*
- [17] Zhihang Zhong, Gurunandan Krishnan, Xiao Sun, Yu Qiao, **Sizhuo Ma***, Jian Wang*. Clearer Frames, Anytime: Resolving Velocity Ambiguity in Video Frame Interpolation. *European Conference on Computer Vision (ECCV 2024, Oral)*
- [18] Dasong Li, Wenjie Li, Baili Lu, Hongsheng Li, **Sizhuo Ma**, Gurunandan Krishnan, Jian Wang*. Delving Deep into Engagement Prediction of Short Videos. *European Conference on Computer Vision (ECCV 2024)*
- [19] Dorian Chan, Matthew O’Toole, **Sizhuo Ma**, Jian Wang. Holodepth: Programmable Depth-Varying Projection via Computer-Generated Holography. *European Conference on Computer Vision (ECCV 2024)*
- [20] Zhixiang Wang, Yu-Lun Liu, Jia-Bin Huang, Shin’ichi Satoh, **Sizhuo Ma**, Guru Krishnan, Jian Wang*. DisCO:

Portrait Distortion Correction with Perspective-Aware 3D GANs. *International Journal of Computer Vision (IJCV 2024)*

- [21] Wei-Ting Chen, Gurunandan Krishnan, Qiang Gao, Sy-Yen Kuo, **Sizhuo Ma***, Jian Wang*+. DSL-FIQA: Assessing Facial Image Quality via Dual-Set Degradation Learning and Landmark-Guided Transformer. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024)*
- [22] Wei-Ting Chen, Yu-Jiet Vong, Sy-Yen Kuo, **Sizhuo Ma***, Jian Wang*. RobustSAM: Segment Anything Robustly on Degraded Images. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024, Highlight)*
- [23] **Sizhuo Ma**, Varun Sundar, Paul Mos, Claudio Bruschini, Edoardo Charbon, Mohit Gupta. Seeing Photons in Color. *SIGGRAPH 2023 / ACM Transactions on Graphics*
- [24] **Sizhuo Ma**, Jian Wang, Wenzheng Chen, Suman Banerjee, Mohit Gupta, Shree Nayar. QfaR: Location-Guided Scanning of Visual Codes from Long Distances. *International Conference on Mobile Computing and Networking (MobiCom 2023)*
- [25] Rui Yu, Jian Wang, **Sizhuo Ma**, Sharon X. Huang, Gurunandan Krishnan, Yicheng Wu. Be Real in Scale: Swing for True Scale in Dual Camera Mode. *International Symposium on Mixed and Augmented Reality (ISMAR 2023)*
- [26] Brevin Tilmon, Zhanghao Sun, Sanjeev Jagannatha Koppal, Yicheng Wu, Georgios Evangelidis, Ramzi Zahreddine, Gurunandan Krishnan, **Sizhuo Ma***, Jian Wang*. Energy-Efficient Adaptive 3D Sensing. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2023)*
- [27] **Sizhuo Ma**, Paul Mos, Edoardo Charbon, Mohit Gupta. Burst Vision Using Single-Photon Cameras. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2023)*
- [28] Varun Sundar, **Sizhuo Ma**, Aswin Sankarnarayanan, Mohit Gupta. Single-Photon Structured Light. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2022)*
- [29] **Sizhuo Ma**, Mohit Gupta. Inertial Safety from Structured Light. *European Conference on Computer Vision (ECCV 2020)*
- [30] **Sizhuo Ma**, Shantanu Gupta, Arin C. Ulku, Claudio Bruschini, Edoardo Charbon, Mohit Gupta. Quanta Burst Photography. *SIGGRAPH 2020 / ACM Transactions on Graphics*
- [31] **Sizhuo Ma**, Brandon M. Smith, Mohit Gupta. Differential Scene Flow from Light Field Gradients. *International Journal on Computer Vision (IJCV) Special Issue on Best Papers of ECCV 2018*
- [32] **Sizhuo Ma**, Brandon M. Smith, Mohit Gupta. 3D Scene Flow from 4D Light Field Gradients. *European Conference on Computer Vision (ECCV 2018)* [Oral presentation]

Patents

Real-time Selfie Perspective Undistortion on Mobiles by Im2Im Translation

Inventors: Jian Wang, Haiwei Chen, Sizhuo Ma, Gurunandan Krishnan Gorumkonda, US 20260004410 A1

Object Scale Utilizing Away-Facing Images

Inventors: Gurunandan Krishnan Gorumkonda, Rui Yu, Yicheng Wu, Jian Wang, Sizhuo Ma, US 20250238994 A1

AR Mirror

Inventors: Shree K. Nayar, Gurunandan Krishnan Gorumkonda, Jian Wang, Bing Zhou, Sizhuo Ma, Karl Bayer, Yicheng Wu, US 20240355239 A1

Energy-Efficient Adaptive 3D Sensing

Inventors: Jian Wang, Sizhuo Ma, Brevin Tilmon, Yicheng Wu, Gurunandan Krishnan Gorumkonda, Ramzi Zahreddine, Georgios Evangelidis, **Patent granted**, US Patent 12001024

Location-Guided Scanning of Visual Codes

Inventors: Sizhuo Ma, Jian Wang, Mohit Gupta, Shree K. Nayar, US 2022/0262089 A1

Systems, Methods, and Media for High Dynamic Range Quanta Burst Imaging

Inventors: Mohit Gupta, Sizhuo Ma, **Patent granted**, US Patent 11170549 and 11721054

Systems, Methods, and Media for Determining Object Motion in Three Dimensions from Light Field Image Data

Inventors: Mohit Gupta, Sizhuo Ma, Brandon Smith, **Patent granted**, US Patent 10706564

Honors and Awards

2022	Outstanding Graduate-Student Research Award, UW–Madison Computer Sciences Department
2020	Snap Research Fellowship
2012	Shanghai Municipal Scholarship
2011 – 2012	SJTU Academic Excellence Scholarship

Professional Services

Reviewer.

International Conference on Computer Vision (ICCV), 2021, 2023, 2025.

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, 2023, 2024, 2025, 2026.

European Conference on Computer Vision (ECCV), 2022, 2024, 2026.

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2023, 2026.

International Conference on Computational Photography (ICCP), 2023, 2024, 2025, 2026.

AAAI Conference on Artificial Intelligence (AAAI), 2026.

SIGGRAPH, 2024, 2026.

IEEE Transactions on Computational Imaging (TCI).

Workshop Organizer and Host.

Visual Quality Assessment Competition (VQualA) Workshop at ICCV 2025. <https://vquala.github.io/>